



New York State Standards Alignment

For Grades K-8

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About Zearn

Zearn is the 501(c)(3) nonprofit educational organization behind Zearn Math, the top-rated math learning platform used by 1 in 4 elementary-school students and by more than 1 million middle-school students nationwide. Zearn Math supports teachers with research-backed curriculum and digital lessons proven to double student academic growth across all levels of student proficiency. Zearn Math can be used flexibly to support a range of instructional needs including Tier 1 and 2 support, acceleration, intervention, tutoring, after school programming and summer programming.

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Zearn Math is aligned to New York's Learning Standards for Mathematics.

We are committed to ensuring that our high-quality instructional materials align to New York's Learning Standards for Mathematics to fully support New York's students and teachers. The tables on the following pages, organized by New York Learning Standards, identify the alignment between each grade-level standard and Zearn Math lesson(s).

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Kindergarten

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
Counting & Cardinality		
K.CC.1	Count to 100 by ones and by tens.	Mission 5, Topic C, Lesson 13 Mission 5, Topic D, Lesson 15-18
K.CC.2	Count to 100 by ones beginning from any given number (instead of beginning at 1).	Mission 5, Topic C, Lesson 1, 12 Mission 5, Topic D, Lesson 16-18
K.CC.3	Write numbers from 0 to 20. Represent a number of objects with a written numeral 0-20 (with 0 representing a count of no objects).	Mission 1, Topic D, Lesson 12-16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 24, 26-28 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-35, 37 Mission 4, Topic A, Lesson 1-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic E, Lesson 25, 27 Mission 5, Topic A, Lesson 1-4 Mission 5, Topic B, Lesson 6, 9 Mission 5, Topic C, Lesson 10-13
K.CC.4	Understand the relationship between numbers and quantities up to 20; connect counting to cardinality. a. When counting objects, say the number names in the standard order, pairing each object with one and only one number name and each number name with one and only one object. (1:1 correspondence) b. Understand that the last number name said tells the number of objects counted, (cardinality). The number of objects is the same regardless of their arrangement or the order in which they were counted. c. Understand the concept that each successive number name refers to a quantity that is one larger. d. Understand the concept of ordinal numbers (first through tenth) to describe the relative position and magnitude of whole numbers.	Mission 1, Topic B, Lesson 5, 6 Mission 1, Topic C, Lesson 7-11 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 23-28 Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-37 Mission 4, Topic H, Lesson 38

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
K.CC.5	a. Answer counting questions using as many as 20 objects arranged in a line, a rectangular array, and a circle. Answer counting questions using as many as 10 objects in a scattered configuration. b. Given a number from 1–20, count out that many objects.	Mission 1, Topic C, Lesson 8-11 Mission 1, Topic D, Lesson 12-16 Mission 1, Topic E, Lesson 17-22 Mission 1, Topic F, Lesson 23-28 Mission 1, Topic G, Lesson 29, 31-32 Mission 1, Topic H, Lesson 33-37 Mission 3, Topic F, Lesson 23-24 Mission 5, Topic A, Lesson 1-5 Mission 5, Topic B, Lesson 6 Mission 5, Topic C, Lesson 10-14 Mission 5, Topic D, Lesson 18, 19
K.CC.6	Identify whether the number of objects in one group is greater than (more than), less than (fewer than), or equal to (the same as) the number of objects in another group.	Mission 3, Topic E, Lesson 17-19 Mission 3, Topic F, Lesson 20-24 Mission 3, Topic G, Lesson 25-28
K.CC.7	Compare two numbers between 1 and 10 presented as written numerals.	Mission 3, Topic F, Lesson 20 Mission 3, Topic G, Lesson 25-28
Operations & Algebraic Thinking		
K.OA.1	Represent addition and subtraction using objects, fingers, pennies, drawings, sounds, acting out situations, verbal explanations, expressions, equations, or other strategies.	Mission 4, Topic C, Lesson 13-16 Mission 4, Topic D, Lesson 20-24 Mission 4, Topic F, Lesson 29-30 Mission 4, Topic G, Lesson 33 Mission 4, Topic H, Lesson 38
K.OA.2	a. Add and subtract within 10. b. Solve addition and subtraction word problems within 10.	Mission 4, Topic C, Lesson 16-18 Mission 4, Topic D, Lesson 19, 21 Mission 4, Topic F, Lesson 31 Mission 4, Topic G, Lesson 34-36 Mission 4, Topic H, Lesson 37
K.OA.3	Decompose numbers less than or equal to 10 into pairs in more than one way. Record each decomposition with a drawing or equation.	Mission 1, Topic C, Lesson 9-11 Mission 1, Topic D, Lesson 14-15 Mission 4, Topic A, Lesson 2, 4-6 Mission 4, Topic B, Lesson 7-12 Mission 4, Topic D, Lesson 22-24 Mission 4, Topic E, Lesson 25-28 Mission 4, Topic F, Lesson 32

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
K.OA.4	Find the number that makes 10 when given a number from 1 to 9. Record the answer with a drawing or equation.	Mission 4, Topic H, Lesson 39-40
K.OA.5	Fluently add and subtract within 5.	Mission 4, Topic G, Lesson 34-36 Mission 4, Topic H, Lesson 37, 39 Mission 6, Topic A, Lesson 1-2 Mission 6, Topic B, Lesson 5
K.OA.6	Duplicate, extend, and create simple patterns using concrete objects.	Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-36 Mission 4, Topic B, Lesson 10-12 Mission 5, Topic A, Lesson 2, 4-5 Mission 5, Topic B, Lesson 6-7 Mission 5, Topic C, Lesson 11-12
Numbers & Operations in Base 10		
K.NBT.1	Compose and decompose the numbers from 11 to 19 into ten ones and one, two, three, four, five, six, seven, eight, or nine ones.	Mission 5, Topic B, Lesson 6-9 Mission 5, Topic C, Lesson 10-13 Mission 5, Topic E, Lesson 20-21, 23
Measurement & Data		
K.MD.1	Describe measurable attributes of an object(s), such as length or weight, using appropriate vocabulary.	Mission 3, Topic A, Lesson 1-2 Mission 3, Topic C, Lesson 8 Mission 3, Topic D, Lesson 13 Mission 3, Topic E, Lesson 16
K.MD.2	Directly compare two objects with a common measurable attribute and describe the difference.	Mission 3, Topic A, Lesson 1-3 Mission 3, Topic B, Lesson 4-7 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 13-15 Mission 3, Topic F, Lesson 20 Mission 3, Topic G, Lesson 28 Mission 3, Topic H, Lesson 29-32
K.MD.3	Classify objects into given categories; count the objects in each category and sort the categories by count.	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7 Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 7

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
Kindergarten Standards		Lessons
K.MD.4	Explore coins (pennies, nickels, dimes, and quarters) and begin identifying pennies and dimes.	Mission 1, Topic G, Lesson 29-32 Mission 1, Topic H, Lesson 33-36 Mission 4, Topic B, Lesson 10-12 Mission 5, Topic A, Lesson 2, 4-5 Mission 5, Topic B, Lesson 6-7 Mission 5, Topic C, Lesson 11-12
Geometry		
K.G.1	Describe objects in the environment using names of shapes, and describe the relative positions of these objects using terms such as above, below, beside, in front of, behind, and next to.	Mission 2, Topic A, Lesson 5 Mission 2, Topic B, Lesson 8 Mission 2, Topic C, Lesson 10
K.G.2	Name shapes regardless of their orientation or overall size.	Mission 2, Topic A, Lesson 2-5 Mission 2, Topic B, Lesson 7-8 Mission 2, Topic C, Lesson 9-10
K.G.3	Understand the difference between two-dimensional (lying in a plane, "flat") and three-dimensional ("solid") shapes.	Mission 2, Topic A, Lesson 1 Mission 2, Topic B, Lesson 6 Mission 2, Topic C, Lesson 9-10
K.G.4	Analyze, compare, and sort two- and three-dimensional shapes, in different sizes and orientations, using informal language to describe their similarities, differences, parts, and other attributes.	Mission 2, Topic A, Lesson 1-4 Mission 2, Topic B, Lesson 6-7 Mission 2, Topic C, Lesson 9-10 Mission 6, Topic A, Lesson 3
K.G.5	Model objects in their environment by building and/or drawing shapes.	Mission 6, Topic A, Lesson 1-3
K.G.6	Compose larger shapes from simple shapes.	Mission 6, Topic B, Lesson 5-7

1st Grade

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
Operations & Algebraic Thinking		
1.OA.1	Use addition and subtraction within 20 to solve one step word problems involving situations of adding to, taking from, putting together, taking apart, and/or comparing, with unknowns in all positions.	Mission 1, Topic B, Lesson 4-8 Mission 1, Topic C, Lesson 9-13 Mission 1, Topic G, Lesson 25 Mission 1, Topic H, Lesson 28-32 Mission 2, Topic A, Lesson 3-5, 7-9, 11 Mission 2, Topic B, Lesson 12-13, 17-18, 21 Mission 2, Topic C, Lesson 22-24 Mission 2, Topic D, Lesson 27-29 Mission 3, Topic D, Lesson 10-13 Mission 4, Topic E, Lesson 19-22 Mission 6, Topic A, Lesson 1-2
1.OA.2	Solve word problems that call for addition of three whole numbers whose sum is less than or equal to 20.	Mission 2, Topic A, Lesson 1
1.OA.3	Apply properties of operations as strategies to add and subtract.	Mission 1, Topic E, Lesson 19, 20 Mission 1, Topic F, Lesson 21-24 Mission 2, Topic A, Lesson 2, 6, 10-11
1.OA.4	Understand subtraction as an unknown-addend problem within 20.	Mission 1, Topic G, Lesson 26, 27 Mission 1, Topic H, Lesson 29-30, 32 Mission 2, Topic B, Lesson 14-19, 21 Mission 2, Topic C, Lesson 22
1.OA.5	Relate counting to addition and subtraction.	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 4-8 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic G, Lesson 26, 27 Mission 2, Topic A, Lesson 5, 9 Mission 2, Topic B, Lesson 12-17, 19, 21

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
1.OA.6	a. Add and subtract within 20. Use strategies such as: counting on, making ten, decomposing a number leading to a ten, using the relationship between addition and subtraction, and creating equivalent but easier or known sums. b. Fluently add and subtract within 10.	Mission 1, Topic C, Lesson 9-13 Mission 1, Topic D, Lesson 14-16 Mission 1, Topic E, Lesson 17-20 Mission 1, Topic F, Lesson 21-24 Mission 1, Topic G, Lesson 26-27 Mission 1, Topic I, Lesson 33-37 Mission 1, Topic J, Lesson 38-39 Mission 2, Topic A, Lesson 1-11 Mission 2, Topic B, Lesson 12-21 Mission 2, Topic C, Lesson 25 Mission 2, Topic D, Lesson 27-29
1.OA.7	Understand the meaning of the equal sign, and determine if equations involving addition and subtraction are true or false.	Mission 1, Topic E, Lesson 17-19 Mission 2, Topic A, Lesson 6, 10 Mission 2, Topic B, Lesson 20 Mission 2, Topic C, Lesson 25
1.OA.8	Determine the unknown whole number in an addition or subtraction equation with the unknown in all positions.	Mission 1, Topic D, Lesson 16 Mission 1, Topic H, Lesson 29-30, 32 Mission 2, Topic A, Lesson 10 Mission 2, Topic B, Lesson 20
Numbers & Operations in Base 10		
1.NBT.1	Count to 120, starting at any number less than 120. In this range, read and write numerals and represent a number of objects with a written numeral.	Mission 4, Topic A, Lesson 1-6 Mission 6, Topic B, Lesson 4, 7, 9
1.NBT.2	Understand that the two digits of a two-digit number represent amounts of tens and ones. a. Understand 10 can be thought of as a bundle of ten ones, called a "ten". b. Understand the numbers from 11 to 19 are composed of a ten and one, two, three, four, five, six, seven, eight, or nine ones. c. Understand the numbers 10, 20, 30, 40, 50, 60, 70, 80, 90 refer to one, two, three, four, five, six, seven, eight, or nine tens (and 0 ones).	Mission 2, Topic D, Lesson 26-29 Mission 4, Topic A, Lesson 1-4, 6 Mission 4, Topic F, Lesson 23 Mission 6, Topic B, Lesson 3-4, 8-9
1.NBT.3	Compare two two-digit numbers based on meanings of the tens and ones digits, recording the results of comparisons with the symbols $>$, $=$, and $<$.	Mission 4, Topic B, Lesson 7-10 Mission 6, Topic B, Lesson 6

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
1.NBT.4	Add within 100, including a two-digit number and a one-digit number, a two-digit number and a multiple of 10. Use concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction. Understand that in adding two-digit numbers, one adds tens and tens, ones and ones, and sometimes it is necessary to compose a ten. Relate the strategy to a written representation and explain the reasoning used.	Mission 4, Topic C, Lesson 11-12 Mission 4, Topic D, Lesson 13-18 Mission 4, Topic F, Lesson 23-29 Mission 6, Topic C, Lesson 10-17 Mission 6, Topic D, Lesson 18-19
1.NBT.5	Given a two-digit number, mentally find 10 more or 10 less than the number, without having to count; explain the reasoning used.	Mission 4, Topic A, Lesson 5-6 Mission 6, Topic B, Lesson 5
1.NBT.6	Subtract multiples of 10 from multiples of 10 in the range 10-90 using concrete models or drawings, and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction. Relate the strategy used to a written representation and explain the reasoning.	Mission 4, Topic C, Lesson 11 Mission 6, Topic C, Lesson 10
Measurement & Data		
1.MD.1	Order three objects by length; compare the lengths of two objects indirectly by using a third object.	Mission 3, Topic A, Lesson 1-3 Mission 3, Topic B, Lesson 5-6 Mission 3, Topic C, Lesson 9
1.MD.2	Measure the length of an object using same-size "length units" placed end to end with no gaps or overlaps. Express the length of an object as a whole number of "length units."	Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 7-9
1.MD.3	a. Tell and write time in hours and half-hours using analog and digital clocks. Develop an understanding of common terms, such as, but not limited to, o'clock and half past. b. Recognize and identify coins (penny, nickel, dime, and quarter) and their value and use the cent symbol (¢) appropriately. c. Count a mixed collection of dimes and pennies and determine the cent value (total not to exceed 100 cents).	Mission 5, Topic D, Lesson 10-13 Mission 6, Topic E, Lesson 20-24

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
1st Grade Standards		Lessons
1.MD.4	Organize, represent, and interpret data with up to three categories; ask and answer questions about the total number of data points, how many in each category, and how many more or less are in one category than in another.	Mission 3, Topic D, Lesson 10-13
Geometry		
1.G.1	Distinguish between defining attributes versus non-defining attributes for a wide variety of shapes. Build and/or draw shapes to possess defining attributes.	Mission 5, Topic A, Lesson 1-3
1.G.2	Compose two-dimensional shapes (rectangles, squares, trapezoids, triangles, half-circles, and quarter-circles) or three-dimensional shapes (cubes, right rectangular prisms, right circular cones, and right circular cylinders) to create a composite shape, and compose new shapes from the composite shape.	Mission 5, Topic B, Lesson 4-6
1.G.3	Partition circles and rectangles into two and four equal shares, describe the shares using the words halves, fourths, and quarters, and use the phrases half of, fourth of, and quarter of. Describe the whole as two of, or four of the shares. Understand for these examples that decomposing into more equal shares creates smaller shares.	Mission 5, Topic C, Lesson 7-9 Mission 5, Topic D, Lesson 10

2nd Grade

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
Operations & Algebraic Thinking		
2.OA.1	a. Use addition and subtraction within 100 to solve one-step word problems involving situations of adding to, taking from, putting together, taking apart, and comparing, with unknowns in all positions. b. Use addition and subtraction within 100 to develop an understanding of solving two-step word problems involving situations of adding to, taking from, putting together, taking apart, and comparing, with unknowns in all positions.	Mission 1, Topic B, Lesson 3, 5-6, 8 Mission 4, Topic A, Lesson 1-5 Mission 4, Topic B, Lesson 6-10 Mission 4, Topic C, Lesson 11-14, 16 Mission 4, Topic F, Lesson 31 Mission 7, Topic B, Lesson 7-8, 12-13
2.OA.2	a. Fluently add and subtract within 20 using mental strategies. Strategies could include: counting on, making ten, decomposing a number leading to a ten, using the relationship between addition and subtraction, and creating equivalent but easier or known sums. b. Know from memory all sums within 20 of two one-digit numbers.	Mission 1, Topic A, Lesson 1-2 Mission 1, Topic B, Lesson 4, 7
2.OA.3	a. Determine whether a group of objects (up to 20) has an odd or even number of members. b. Write an equation to express an even number as a sum of two equal addends.	Mission 6, Topic D, Lesson 17-19
2.OA.4	Use addition to find the total number of objects arranged in rectangular arrays with up to 5 rows and up to 5 columns. Write an equation to express the total as a sum of equal addends.	Mission 6, Topic A, Lesson 1-4 Mission 6, Topic B, Lesson 5-9 Mission 6, Topic C, Lesson 10-15
Numbers & Operations in Base 10		
2.NBT.1	Understand that the digits of a three-digit number represent amounts of hundreds, tens, and ones. a. Understand 100 can be thought of as a bundle of ten tens, called a "hundred." b. Understand the numbers 100, 200, 300, 400, 500, 600, 700, 800, 900 refer to one, two, three, four, five, six, seven, eight, or nine hundreds (and 0 tens and 0 ones).	Mission 3, Topic A, Lesson 1 Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 11-15

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
2.NBT.2	Count within 1000; skip-count by 5s, 10s, and 100s.	Mission 3, Topic B, Lesson 2-3 Mission 3, Topic C, Lesson 4 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 12-15 Mission 3, Topic G, Lesson 19-21 Mission 7, Topic E, Lesson 21 Mission 8, Topic D, Lesson 14
2.NBT.3	Read and write numbers to 1000 using base-ten numerals, number names, and expanded form.	Mission 3, Topic C, Lesson 4-7 Mission 3, Topic D, Lesson 8-9 Mission 3, Topic E, Lesson 13, 14
2.NBT.4	Compare two three-digit numbers based on meanings of the hundreds, tens, and ones digits, using $>$, $=$, and $<$ symbols to record the results of comparisons.	Mission 3, Topic F, Lesson 16-18
2.NBT.5	Fluently add and subtract within 100 using strategies based on place value, properties of operations, and/or the relationship between addition and subtraction.	Mission 1, Topic B, Lesson 3, 5-6, 8 Mission 4, Topic A, Lesson 2-4 Mission 4, Topic B, Lesson 6-8, 10 Mission 4, Topic C, Lesson 11-13 Mission 7, Topic B, Lesson 7-8, 11-13
2.NBT.6	Add up to four two-digit numbers using strategies based on place value and properties of operations.	Mission 4, Topic D, Lesson 22
2.NBT.7	a. Add and subtract within 1000, using concrete models or drawings, and strategies based on place value, properties of operations, and/or the relationship between addition and subtraction. Relate the strategy to a written representation. b. Understand that in adding or subtracting up to three-digit numbers, one adds or subtracts hundreds and hundreds, tens and tens, ones and ones, and sometimes it is necessary to compose or decompose tens or hundreds.	Mission 4, Topic B, Lesson 9 Mission 4, Topic C, Lesson 14-15 Mission 4, Topic D, Lesson 17-21 Mission 4, Topic E, Lesson 23-28 Mission 4, Topic F, Lesson 29-30 Mission 5, Topic A, Lesson 1-7 Mission 5, Topic B, Lesson 8-12 Mission 5, Topic C, Lesson 13-18 Mission 5, Topic D, Lesson 19-20
2.NBT.8	Mentally add 10 or 100 to a given number 100-900, and mentally subtract 10 or 100 from a given number 100-900.	Mission 3, Topic G, Lesson 19-20 Mission 4, Topic D, Lesson 17 Mission 5, Topic A, Lesson 1-2
2.NBT.9	Explain why addition and subtraction strategies work, using place value and the properties of operations.	Mission 4, Topic C, Lesson 11 Mission 4, Topic F, Lesson 30

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
		Mission 5, Topic A, Lesson 7 Mission 5, Topic B, Lesson 12 Mission 5, Topic C, Lesson 13-18 Mission 5, Topic D, Lesson 19-20
Measurement & Data		
2.MD.1	Measure the length of an object to the nearest whole by selecting and using appropriate tools such as rulers, yardsticks, meter sticks, and measuring tapes.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 4-5 Mission 2, Topic C, Lesson 6 Mission 7, Topic C, Lesson 14-15 Mission 7, Topic D, Lesson 16-17
2.MD.2	Measure the length of an object twice, using different "length units" for the two measurements; describe how the two measurements relate to the size of the unit chosen.	Mission 2, Topic C, Lesson 7 Mission 7, Topic D, Lesson 18
2.MD.3	Estimate lengths using units of inches, feet, centimeters, and meters.	Mission 2, Topic B, Lesson 5 Mission 2, Topic D, Lesson 9 Mission 7, Topic D, Lesson 17
2.MD.4	Measure to determine how much longer one object is than another, expressing the length difference in terms of a standard "length unit."	Mission 2, Topic C, Lesson 6-7 Mission 2, Topic D, Lesson 9 Mission 7, Topic D, Lesson 19
2.MD.5	Use addition and subtraction within 100 to solve word problems involving lengths that are given in the same units.	Mission 2, Topic D, Lesson 8-10 Mission 7, Topic E, Lesson 20, 22
2.MD.6	Represent whole numbers as lengths from 0 on a number line with equally spaced points corresponding to the numbers 0, 1, 2, ..., and represent whole-number sums and differences within 100 on a number line.	Mission 2, Topic D, Lesson 8 Mission 7, Topic A, Lesson 3 Mission 7, Topic E, Lesson 21-22 Mission 7, Topic F, Lesson 24
2.MD.7	Tell and write time from analog and digital clocks in five minute increments, using a.m. and p.m. Develop an understanding of common terms, such as, but not limited to, quarter past, half past, and quarter to.	Mission 8, Topic D, Lesson 13-15
2.MD.8	a. Count a mixed collection of coins whose sum is less than or equal to one dollar.	Mission 3, Topic D, Lesson 10

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
2nd Grade Standards		Lessons
	b. Solve real world and mathematical problems within one dollar involving quarters, dimes, nickels, and pennies, using the ¢ (cent) symbol appropriately.	Mission 7, Topic B, Lesson 6-10, 12-13
2.MD.9	Generate measurement data by measuring lengths of several objects to the nearest whole unit, or by making repeated measurements of the same object. Present the measurement data in a line plot, where the horizontal scale is marked off in whole-number units.	Mission 7, Topic F, Lesson 23-26
2.MD.10	Draw a picture graph and a bar graph (with single-unit scale) to represent a data set with up to four categories. Solve simple put-together, take-apart, and compare problems using information presented in a picture graph or a bar graph.	Mission 7, Topic A, Lesson 1-5
Geometry		
2.G.1	Classify two-dimensional figures as polygons or non-polygons.	Mission 8, Topic A, Lesson 1-4 Grade 3 Mission 7, Topic B, Lesson 4-6
2.G.2	Partition a rectangle into rows and columns of same-size squares and count to find the total number of them.	Mission 6, Topic C, Lesson 14-15
2.G.3	Partition circles and rectangles into two, three, or four equal shares. Describe the shares using the words halves, thirds, half of, a third of, etc. Describe the whole as two halves, three thirds, four fourths. Recognize that equal shares of identical wholes need not have the same shape.	Mission 8, Topic B, Lesson 7-8 Mission 8, Topic C, Lesson 9-12 Mission 8, Topic D, Lesson 13

3rd Grade

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
Operations & Algebraic Thinking		
3.OA.1	Interpret products of whole numbers.	Mission 1, Topic A, Lesson 1-3 Mission 1, Topic B, Lesson 6 Mission 1, Topic C, Lesson 7-10 Mission 1, Topic E, Lesson 15, 17 Mission 3, Topic D, Lesson 12 Mission 3, Topic E, Lesson 16
3.OA.2	Interpret whole-number quotients of whole numbers.	Mission 1, Topic B, Lesson 4-6 Mission 1, Topic D, Lesson 12-13 Mission 1, Topic E, Lesson 17
3.OA.3	Use multiplication and division within 100 to solve word problems in situations involving equal groups, arrays, and measurement quantities.	Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 8-9 Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14-17 Mission 1, Topic F, Lesson 18 Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 7 Mission 3, Topic C, Lesson 11 Mission 3, Topic D, Lesson 15
3.OA.4	Determine the unknown whole number in a multiplication or division equation relating three whole numbers.	Mission 1, Topic B, Lesson 6 Mission 1, Topic C, Lesson 8 Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lesson 17 Mission 3, Topic A, Lesson 3 Mission 3, Topic D, Lesson 13 Mission 3, Topic E, Lesson 16
3.OA.5	Apply properties of operations as strategies to multiply and divide.	Mission 1, Topic C, Lesson 7-10 Mission 1, Topic E, Lesson 15-16 Mission 1, Topic F, Lesson 18-19 Mission 3, Topic A, Lesson 1, 2 Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 9-10 Mission 3, Topic D, Lesson 12-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 20

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
3.OA.6	Understand division as an unknown-factor problem.	Mission 1, Topic B, Lesson 6 Mission 1, Topic D, Lesson 11 Mission 1, Topic E, Lesson 17 Mission 3, Topic D, Lesson 12, 13 Mission 3, Topic E, Lesson 16
3.OA.7	a. Fluently solve single-digit multiplication and related divisions, using strategies such as the relationship between multiplication and division or properties of operations. b. Know from memory all products of two one-digit numbers.	Mission 1, Topic D, Lesson 11-13 Mission 1, Topic E, Lesson 14, 16 Mission 3, Topic B, Lesson 4-5 Mission 3, Topic C, Lesson 10 Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
3.OA.8	Solve two-step word problems posed with whole numbers and having whole-number answers using the four operations. a. Represent these problems using equations or expressions with a letter standing for the unknown quantity. b. Assess the reasonableness of answers using mental computation and estimation strategies including rounding.	Mission 1, Topic F, Lesson 20-21 Mission 3, Topic E, Lesson 18 Mission 3, Topic F, Lesson 21 Mission 7, Topic A, Lesson 1-3
3.OA.9	Identify and extend arithmetic patterns (including patterns in the addition table or multiplication table).	Mission 3, Topic A, Lesson 1 Mission 3, Topic D, Lesson 13-14 Mission 3, Topic E, Lesson 16-17 Mission 3, Topic F, Lesson 19
Numbers & Operations in Base 10		
3.NBT.1	Use place value understanding to round whole numbers to the nearest 10 or 100.	Mission 2, Topic C, Lesson 12-14 Mission 2, Topic D, Lesson 17 Mission 2, Topic E, Lesson 20-21
3.NBT.2	Fluently add and subtract within 1,000 using strategies and algorithms based on place value, properties of operations, and/or the relationship between addition and subtraction.	Mission 2, Topic A, Lesson 4-5 Mission 2, Topic B, Lesson 10-11 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21
3.NBT.3	Multiply one-digit whole numbers by multiples of 10 in the range 10-90 using strategies based on place value and properties of operations.	Mission 3, Topic F, Lesson 19-20

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
3.NBT.4	a. Understand that the digits of a four-digit number represent amounts of thousands, hundreds, tens, and ones. b. Read and write four-digit numbers using base-ten numerals, number names, and expanded form.	Grade 3 "Form to Form" fluency activities included in Independent Digital Lessons cover the full extent of the standard
Numbers & Operations - Fractions		
3.NF.1	Understand a unit fraction, $\frac{1}{b}$, is the quantity formed by 1 part when a whole is partitioned into b equal parts. Understand a fraction $\frac{a}{b}$ as the quantity formed by a parts of size $\frac{1}{b}$.	Mission 5, Topic B, Lesson 5-9 Mission 5, Topic C, Lesson 10, 12-13 Mission 5, Topic E, Lesson 20, 22, 24-25, 27 Mission 5, Topic F, Lesson 28-29
3.NF.2	Understand a fraction as a number on the number line; represent fractions on a number line. a. Represent a fraction $\frac{1}{b}$ on a number line by defining the interval from 0 to 1 as the whole and partitioning it into b equal parts. Recognize that each part has size $\frac{1}{b}$ and that the endpoint of the part starting at 0 locates the number $\frac{1}{b}$ on the number line. b. Represent a fraction $\frac{a}{b}$ on a number line by marking off a lengths $\frac{1}{b}$ from 0. Recognize that the resulting interval has size $\frac{a}{b}$ and that its endpoint locates the number $\frac{a}{b}$ on the number line.	Mission 5, Topic D, Lesson 14-17, 19 Mission 5, Topic E, Lesson 21, 23, 26
3.NF.3	Explain equivalence of fractions and compare fractions by reasoning about their size. a. Understand two fractions as equivalent (equal) if they are the same size, or the same point on a number line. b. Recognize and generate equivalent fractions. Explain why the fractions are equivalent. c. Express whole numbers as fractions, and recognize fractions that are equivalent to whole numbers. d. Compare two fractions with the same numerator or the same denominator by reasoning about their size. Recognize that comparisons rely on the two fractions referring to the same whole. Record the results of comparisons with the symbols $>$, $=$, or $<$, and justify the conclusions.	Mission 5, Topic C, Lesson 10-11, 13 Mission 5, Topic D, Lesson 16-19 Mission 5, Topic E, Lesson 20-27 Mission 5, Topic F, Lesson 28-29
Measurement & Data		

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
3.MD.1	Tell and write time to the nearest minute and measure time intervals in minutes. Solve one-step word problems involving addition and subtraction of time intervals in minutes.	Mission 2, Topic A, Lesson 1-5 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18, 21
3.MD.2	a. Measure and estimate liquid volumes and masses of objects using grams (g), kilograms (kg), and liters (l). b. Add, subtract, multiply, or divide to solve one-step word problems involving masses or liquid volumes that are given in the same units.	Mission 2, Topic B, Lesson 6-11 Mission 2, Topic C, Lesson 12 Mission 2, Topic D, Lesson 15-17 Mission 2, Topic E, Lesson 18-21
3.MD.3	Draw a scaled picture graph and a scaled bar graph to represent a data set with several categories. Solve one- and two-step "how many more" and "how many less" problems using information presented in a scaled picture graph or a scaled bar graph.	Mission 6, Topic A, Lesson 1-4
3.MD.4	Generate measurement data by measuring lengths using rulers marked with halves and fourths of an inch. Show the data by making a line plot where the horizontal scale is marked off in appropriate units—whole numbers, halves, or quarters.	Mission 6, Topic B, Lesson 5-9 Mission 7, Topic D, Lesson 19, 22 Mission 7, Topic E, Lesson 26
3.MD.5	Recognize area as an attribute of plane figures and understand concepts of area measurement. a. Recognize a square with side length 1 unit, called "a unit square," is said to have "one square unit" of area, and can be used to measure area. b. Recognize a plane figure which can be covered without gaps or overlaps by n unit squares is said to have an area of n square units.	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic B, Lesson 5-7 Mission 4, Topic D, Lesson 13
3.MD.6	Measure areas by counting unit squares.	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic B, Lesson 5-7 Mission 4, Topic D, Lesson 13
3.MD.7	Relate area to the operations of multiplication and addition. a. Find the area of a rectangle with whole-number side lengths by tiling it, and show that the area is the same as would be found by multiplying the side lengths. b. Multiply side lengths to find areas of rectangles with whole-number side lengths in the context of solving real world and mathematical	Mission 4, Topic A, Lesson 4 Mission 4, Topic B, Lesson 5-8 Mission 4, Topic C, Lesson 9-11 Mission 4, Topic D, Lesson 12-16 Mission 7, Topic E, Lesson 28-30

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
3rd Grade Standards		Lessons
	problems, and represent whole-number products as rectangular areas in mathematical reasoning. c. Use tiling to show in a concrete case that the area of a rectangle with whole-number side length a and side length $b + c$ is the sum of $a \times b$ and $a \times c$. Use area models to represent the distributive property in mathematical reasoning. d. Recognize area as additive. Find areas of figures composed of non-overlapping rectangles, and apply this technique to solve real world problems.	
3.MD.8	a. Solve real world and mathematical problems involving perimeters of polygons, including finding the perimeter given the side lengths or finding one unknown side length given the perimeter and other side lengths. b. Identify rectangles with the same perimeter and different areas or with the same area and different perimeters.	Mission 7, Topic C, Lesson 10-17 Mission 7, Topic D, Lesson 18-22 Mission 7, Topic E, Lesson 23-30
Geometry		
3.G.1	Recognize and classify polygons based on the number of sides and vertices (triangles, quadrilaterals, pentagons, and hexagons). Identify shapes that do not belong to one of the given subcategories.	Mission 7, Topic B, Lesson 4-9
3.G.2	Partition shapes into parts with equal areas. Express the area of each part as a unit fraction of the whole.	Mission 5, Topic A, Lesson 1-4 Mission 5, Topic B, Lesson 5-9 Mission 5, Topic C, Lesson 10-13 Mission 5, Topic E, Lesson 24-25, 27 Mission 5, Topic F, Lesson 28-29

4th Grade

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
Operations & Algebraic Thinking		
4.OA.1	Interpret a multiplication equation as a comparison. Represent verbal statements of multiplicative comparisons as multiplication equations.	Mission 1, Topic A, Lesson 1
4.OA.2	Multiply or divide to solve word problems involving multiplicative comparison, distinguishing multiplicative comparison from additive comparison. Use drawings and equations with a symbol for the unknown number to represent the problem.	Mission 3, Topic A, Lesson 2-3 Mission 3, Topic B, Lesson 5-6 Mission 3, Topic C, Lesson 9-11 Mission 3, Topic D, Lesson 12, 13 Mission 3, Topic G, Lesson 26 Mission 5, Topic G, Lesson 39-40 Mission 7, Topic A, Lesson 4 Mission 7, Topic C, Lesson 14
4.OA.3	Solve multistep word problems posed with whole numbers and having whole-number answers using the four operations, including problems in which remainders must be interpreted. a. Represent these problems using equations or expressions with a letter standing for the unknown quantity. b. Assess the reasonableness of answers using mental computation and estimation strategies including rounding.	Mission 3, Topic D, Lesson 12-13 Mission 3, Topic E, Lesson 14, 19 Mission 3, Topic G, Lesson 31-32
4.OA.4	Find all factor pairs for a whole number in the range 1–100. Recognize that a whole number is a multiple of each of its factors. Determine whether a given whole number in the range 1–100 is a multiple of a given one-digit number. Determine whether a given whole number in the range 1–100 is prime or composite.	Mission 3, Topic F, Lesson 22-25
4.OA.5	Generate a number or shape pattern that follows a given rule. Identify and informally explain apparent features of the pattern that were not explicit in the rule itself.	Mission 5, Topic H, Lesson 41
Numbers & Operations in Base 10		

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.NBT.1	Recognize that in a multi-digit whole number, a digit in one place represents ten times what it represents in the place to its right.	Mission 1, Topic A, Lesson 2-3
4.NBT.2	a. Read and write multi-digit whole numbers using base-ten numerals, number names, and expanded form. b. Compare two multi-digit numbers based on meanings of the digits in each place, using $>$, $=$, and $<$ symbols to record the results of comparisons.	Mission 1, Topic A, Lesson 3-4 Mission 1, Topic B, Lesson 5-6
4.NBT.3	Use place value understanding to round multi-digit whole numbers to any place.	Mission 1, Topic C, Lesson 7-10 Mission 1, Topic D, Lesson 12 Mission 1, Topic E, Lesson 16 Mission 1, Topic F, Lesson 17, 19
4.NBT.4	Fluently add and subtract multi-digit whole numbers using a standard algorithm.	Mission 1, Topic D, Lesson 11-12 Mission 1, Topic E, Lesson 13-16 Mission 1, Topic F, Lesson 17-19
4.NBT.5	Multiply a whole number of up to four digits by a one-digit whole number, and multiply two two-digit numbers, using strategies based on place value and the properties of operations. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 3, Topic B, Lesson 4-6 Mission 3, Topic C, Lesson 7-11 Mission 3, Topic H, Lesson 34-38
4.NBT.6	Find whole-number quotients and remainders with up to four-digit dividends and one-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 3, Topic E, Lesson 14-21 Mission 3, Topic G, Lesson 26-33
Numbers & Operations - Fractions		
4.NF.1	Explain why a fraction a/b is equivalent to a fraction $(a \times n)/(b \times n)$ by using visual fraction models, with attention to how the number and size of the parts differ even though the two fractions themselves are the same size. Use this principle to recognize and generate equivalent fractions.	Mission 5, Topic B, Lesson 7-11 Mission 5, Topic C, Lesson 14-15 Mission 5, Topic D, Lesson 21 Mission 5, Topic E, Lesson 27 Mission 6, Topic B, Lesson 5 Mission 6, Topic D, Lesson 12

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.NF.2	Compare two fractions with different numerators and different denominators. Recognize that comparisons are valid only when the two fractions refer to the same whole. Record the results of comparisons with symbols $>$, $=$, or $<$, and justify the conclusions.	Mission 5, Topic C, Lesson 12-15 Mission 5, Topic E, Lesson 26-27
4.NF.3	Understand a fraction a/b with $a > 1$ as a sum of fractions $1/b$. a. Understand addition and subtraction of fractions as joining and separating parts referring to the same whole. b. Decompose a fraction into a sum of fractions with the same denominator in more than one way, recording each decomposition by an equation. Justify decompositions. c. Add and subtract mixed numbers with like denominators. d. Solve word problems involving addition and subtraction of fractions referring to the same whole and having like denominators.	Mission 5, Topic A, Lesson 1-6 Mission 5, Topic D, Lesson 16-21 Mission 5, Topic E, Lesson 22-25 Mission 5, Topic F, Lesson 29-34
4.NF.4	Apply and extend previous understandings of multiplication to multiply a whole number by a fraction. a. Understand a fraction a/b as a multiple of $1/b$. b. Understand a multiple of a/b as a multiple of $1/b$, and use this understanding to multiply a whole number by a fraction. c. Solve word problems involving multiplication of a whole number by a fraction.	Mission 5, Topic A, Lesson 3, 5-6 Mission 5, Topic E, Lesson 23-25 Mission 5, Topic G, Lesson 35-40
4.NF.5	Express a fraction with denominator 10 as an equivalent fraction with denominator 100, and use this technique to add two fractions with respective denominators 10 and 100.	Mission 6, Topic B, Lesson 4 Mission 6, Topic D, Lesson 12-14
4.NF.6	Use decimal notation for fractions with denominators 10 or 100.	Mission 6, Topic A, Lesson 1-3 Mission 6, Topic B, Lesson 4-6 Mission 6, Topic C, Lesson 11 Mission 6, Topic D, Lesson 12-14 Mission 6, Topic E, Lesson 15
4.NF.7	Compare two decimals to hundredths by reasoning about their size. Recognize that comparisons are valid only when two decimals refer to the same whole. Record the results of comparisons with the symbols $>$, $=$, or $<$, and justify the conclusions.	Mission 6, Topic C, Lesson 9-11

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
Measurement & Data		
4.MD.1	Know relative sizes of measurement units: ft., in.; km, m, cm. Know the conversion factor and use it to convert measurements in a larger unit in terms of a smaller unit: ft., in.; km, m, cm; hr., min., sec. Given the conversion factor, convert all other measurements within a single system of measurement from a larger unit to a smaller unit. Record measurement equivalents in a two-column table.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 4 Mission 4, Topic A, Lesson 4 Mission 7, Topic A, Lesson 1-3 Mission 7, Topic B, Lesson 6-9 Mission 7, Topic C, Lesson 12-13
4.MD.2	Use the four operations to solve word problems involving distances, intervals of time, liquid volumes, masses of objects, and money. a. Solve problems involving fractions or decimals, and problems that require expressing measurements given in a larger unit in terms of a smaller unit. b. Represent measurement quantities using diagrams that feature a measurement scale, such as number lines.	Mission 2, Topic A, Lesson 1-3 Mission 2, Topic B, Lesson 5 Mission 5, Topic G, Lesson 40 Mission 6, Topic D, Lesson 14 Mission 6, Topic E, Lesson 15-16 Mission 7, Topic A, Lesson 1-5 Mission 7, Topic B, Lesson 6-11 Mission 7, Topic C, Lesson 14
4.MD.3	Apply the area and perimeter formulas for rectangles in real world and mathematical problems.	Mission 3, Topic A, Lesson 1-3 Mission 7, Topic D, Lesson 15-16
4.MD.4	Make a line plot to display a data set of measurements in fractions of a unit ($\frac{1}{2}, \frac{1}{4}, \frac{1}{8}$). Solve problems involving addition and subtraction of fractions by using information presented in line plots.	Mission 5, Topic E, Lesson 28 Mission 5, Topic G, Lesson 40
4.MD.5	Recognize angles as geometric shapes that are formed wherever two rays share a common endpoint, and understand concepts of angle measurement. a. Recognize an angle is measured with reference to a circle with its center at the common endpoint of the rays, by considering the fraction of the circular arc between the points where the two rays intersect the circle. An angle that turns through $\frac{1}{360}$ of a circle is called a "one-degree angle," and can be used to measure angles. b. Recognize an angle that turns through n one-degree angles is said to have an angle measure of n degrees.	Mission 4, Topic A, Lesson 1 Mission 4, Topic B, Lesson 5-7
4.MD.6	Measure angles in whole-number degrees using a protractor. Sketch angles of specified measure.	Mission 4, Topic B, Lesson 5-8

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
4th Grade Standards		Lessons
4.MD.7	Recognize angle measure as additive. When an angle is decomposed into non-overlapping parts, the angle measure of the whole is the sum of the angle measures of the parts. Solve addition and subtraction problems to find unknown angles on a diagram in real world and mathematical problems.	Mission 4, Topic C, Lesson 9-11
Geometry		
4.G.1	Draw points, lines, line segments, rays, angles (right, acute, obtuse), and perpendicular and parallel lines. Identify these in two-dimensional figures.	Mission 4, Topic A, Lesson 1-4 Mission 4, Topic D, Lesson 13-15
4.G.2	a. Identify and name triangles based on angle size (right, obtuse, acute). b. Identify and name all quadrilaterals with 2 pairs of parallel sides as parallelograms. c. Identify and name all quadrilaterals with four right angles as rectangles.	Mission 4, Topic D, Lesson 13-15
4.G.3	Recognize a line of symmetry for a two-dimensional figure as a line across the figure such that the figure can be folded along the line into matching parts. Identify line-symmetric figures and draw lines of symmetry.	Mission 4, Topic D, Lesson 12

5th Grade

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
Operations & Algebraic Thinking		
5.OA.1	Apply the order of operations to evaluate numerical expressions.	Mission 2, Topic A, Lesson 1, 3-4 Mission 2, Topic B, Lesson 5 Mission 4, Topic D, Lesson 10 Mission 4, Topic H, Lesson 32
5.OA.2	Write simple expressions that record calculations with numbers, and interpret numerical expressions without evaluating them.	Mission 2, Topic B, Lesson 3-4 Mission 4, Topic D, Lesson 10 Mission 4, Topic H, Lesson 32
5.OA.3	Generate two numerical patterns using two given rules. Identify apparent relationships between corresponding terms. Form ordered pairs consisting of corresponding terms from the two patterns, and graph the ordered pairs on a coordinate plane.	Mission 6, Topic B, Lesson 7
Numbers & Operations in Base 10		
5.NBT.1	Recognize that in a multi-digit number, a digit in one place represents 10 times as much as it represents in the place to its right and $\frac{1}{10}$ of what it represents in the place to its left.	Mission 1, Topic A, Lesson 1-4 Mission 2, Topic A, Lesson 2 Mission 2, Topic E, Lesson 16
5.NBT.2	Use whole-number exponents to denote powers of 10. Explain patterns in the number of zeros of the product when multiplying a number by powers of 10, and explain patterns in the placement of the decimal point when a decimal is multiplied or divided by a power of 10.	Mission 1, Topic A, Lesson 1-4 Mission 2, Topic A, Lesson 1 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic D, Lesson 13-14 Mission 2, Topic G, Lesson 24
5.NBT.3	Read, write, and compare decimals to thousandths. a. Read and write decimals to thousandths using base-ten numerals, number names, and expanded form.	Mission 1, Topic B, Lesson 5-6 Mission 1, Topic D, Lesson 9-10 Mission 1, Topic E, Lesson 11-12 Mission 1, Topic F, Lesson 13-15

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
	b. Compare two decimals to thousandths based on meanings of the digits in each place, using $>$, $=$, and $<$ symbols to record the results of comparisons.	
5.NBT.4	Use place value understanding to round decimals to any place.	Mission 1, Topic C, Lesson 7-8
5.NBT.5	Fluently multiply multi-digit whole numbers using a standard algorithm.	Mission 2, Topic B, Lesson 5-9 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic F, Lesson 20-23
5.NBT.6	Find whole-number quotients of whole numbers with up to four-digit dividends and two-digit divisors, using strategies based on place value, the properties of operations, and/or the relationship between multiplication and division. Illustrate and explain the calculation by using equations, rectangular arrays, and/or area models.	Mission 2, Topic E, Lesson 16-18 Mission 2, Topic F, Lesson 19-23 Mission 2, Topic H, Lesson 28-29 Mission 4, Topic G, Lesson 30-31
5.NBT.7	Using concrete models or drawings and strategies based on place value, properties of operations, and/or the relationship between operations: add and subtract decimals to hundredths, multiply and divide decimals to hundredths. Relate the strategy to a written method and explain the reasoning used.	Mission 1, Topic D, Lesson 9-10 Mission 1, Topic E, Lesson 11-12 Mission 1, Topic F, Lesson 13-16 Mission 2, Topic C, Lesson 10-12 Mission 2, Topic D, Lesson 13-14 Mission 2, Topic G, Lesson 24-27 Mission 2, Topic H, Lesson 28-29 Mission 4, Topic E, Lesson 17-18 Mission 4, Topic G, Lesson 29-31
Numbers & Operations - Fractions		
5.NF.1	Add and subtract fractions with unlike denominators (including mixed numbers) by replacing given fractions with equivalent fractions in such a way as to produce an equivalent sum or difference of fractions with like denominators.	Mission 3, Topic A, Lesson 1-2 Mission 3, Topic B, Lesson 3-6 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 14
5.NF.2	Solve word problems involving addition and subtraction of fractions referring to the same whole, including cases of unlike denominators. Use benchmark fractions and number sense of fractions to estimate mentally and assess the reasonableness of answers.	Mission 3, Topic B, Lesson 3-7 Mission 3, Topic C, Lesson 8-12 Mission 3, Topic D, Lesson 13-15 Mission 4, Topic D, Lesson 10-12

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
5.NF.3	Interpret a fraction as division of the numerator by the denominator ($a/b = a \div b$). Solve word problems involving division of whole numbers leading to answers in the form of fractions or mixed numbers.	Mission 4, Topic B, Lesson 2-5
5.NF.4	Apply and extend previous understandings of multiplication to multiply a fraction by a whole number or a fraction. a. Interpret the product $a/b \times q$ as a parts of a partition of q into b equal parts; equivalently, as the result of a sequence of operations $a \times q \div b$. b. Find the area of a rectangle with fractional side lengths by tiling it with rectangles of the appropriate unit fraction side lengths, and show that the area is the same as would be found by multiplying the side lengths. Multiply fractional side lengths to find areas of rectangles, and represent fraction products as rectangular areas.	Mission 4, Topic C, Lesson 6-9 Mission 4, Topic E, Lesson 13-15, 17-18 Mission 5, Topic C, Lesson 10-15
5.NF.5	Interpret multiplication as scaling (resizing). a. Compare the size of a product to the size of one factor on the basis of the size of the other factor, without performing the indicated multiplication. b. Explain why multiplying a given number by a fraction greater than 1 results in a product greater than the given number (recognizing multiplication by whole numbers greater than 1 as a familiar case). Explain why multiplying a given number by a fraction less than 1 results in a product smaller than the given number. Relate the principle of fraction equivalence $a/b = a/b \times n/n$ to the effect of multiplying a/b by 1.	Mission 4, Topic F, Lesson 21-23 Mission 4, Topic H, Lesson 32
5.NF.6	Solve real world problems involving multiplication of fractions and mixed numbers.	Mission 4, Topic C, Lesson 6-7 Mission 4, Topic D, Lesson 10-12 Mission 4, Topic E, Lesson 13-18 Mission 4, Topic F, Lesson 22-24 Mission 5, Topic C, Lesson 12-15
5.NF.7	Apply and extend previous understandings of division to divide unit fractions by whole numbers and whole numbers by unit fractions. a. Interpret division of a unit fraction by a non-zero whole number, and compute such quotients. b. Interpret division of a whole number by a unit fraction, and compute such quotients. c. Solve real-world problems involving	Mission 4, Topic G, Lesson 25-31 Mission 4, Topic H, Lesson 33

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
division of unit fractions by non-zero whole numbers and division of whole numbers by unit fractions.		
Measurement & Data		
5.MD.1	Convert among different-sized standard measurement units within a given measurement system when the conversion factor is given. Use these conversions in solving multi-step, real world problems.	Mission 1, Topic A, Lesson 4 Mission 2, Topic D, Lesson 13-15 Mission 4, Topic C, Lesson 9 Mission 4, Topic E, Lesson 19-20 Mission 5, Topic B, Lesson 5
5.MD.2	Make a line plot to display a data set of measurements in fractions of a unit ($\frac{1}{2}, \frac{1}{4}, \frac{1}{8}$). Use operations on fractions for this grade to solve problems involving information presented in line plots.	Mission 4, Topic A, Lesson 1 Mission 4, Topic D, Lesson 10
5.MD.3	Recognize volume as an attribute of solid figures and understand concepts of volume measurement. a. Recognize that a cube with side length 1 unit, called a "unit cube," is said to have "one cubic unit" of volume, and can be used to measure volume. b. Recognize that a solid figure which can be packed without gaps or overlaps using n unit cubes is said to have a volume of n cubic units.	Mission 5, Topic A, Lesson 1-3 Mission 5, Topic B, Lesson 5
5.MD.4	Measure volumes by counting unit cubes, using cubic cm, cubic in., cubic ft., and improvised units.	Mission 5, Topic A, Lesson 1-3
5.MD.5	Relate volume to the operations of multiplication and addition and solve real world and mathematical problems involving volume. a. Find the volume of a right rectangular prism with whole-number side lengths by packing it with unit cubes, and show that the volume is the same as would be found by multiplying the edge lengths, equivalently by multiplying the height by the area of the base. b. Apply the formulas $V = l \times w \times h$ and $V = B \times h$ for rectangular prisms to find volumes of right rectangular prisms with whole-number edge lengths in the context of solving real world and mathematical problems. c. Recognize volume as additive. Find volumes of solid figures composed of two non-overlapping right rectangular prisms by adding	Mission 5, Topic B, Lesson 4-9

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
5th Grade Standards		Lessons
the volumes of the non-overlapping parts, applying this technique to solve real world problems.		
Geometry		
5.G.1	Use a pair of perpendicular number lines, called axes, to define a coordinate system, with the intersection of the lines (the origin) arranged to coincide with the 0 on each line and a given point in the plane located by using an ordered pair of numbers, called its coordinates. Understand that the first number indicates how far to travel from the origin in the direction of one axis, and the second number indicates how far to travel in the direction of the second axis, with the convention that the names of the two axes and the coordinates correspond.	Mission 6, Topic A, Lesson 1-6 Mission 6, Topic B, Lesson 7-12 Mission 6, Topic C, Lesson 13-17 Mission 6, Topic D, Lesson 18, 20
5.G.2	Represent real world and mathematical problems by graphing points in the first quadrant of the coordinate plane, and interpret coordinate values of points in the context of the situation.	Mission 6, Topic C, Lesson 13-17 Mission 6, Topic D, Lesson 18-20
5.G.3	Understand that attributes belonging to a category of two-dimensional figures also belong to all subcategories of that category.	Mission 5, Topic D, Lesson 16-19, 21
5.G.4	Classify two-dimensional figures in a hierarchy based on properties.	Mission 5, Topic D, Lesson 16-21

6th Grade

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
Ratios & Proportional Relationships		
6.RP.1	Understand the concept of a ratio and use ratio language to describe a ratio relationship between two quantities.	Mission 2, Topic A, Lesson 1-2 Mission 2, Topic B, Lesson 3-5 Mission 9, Topic A, Lesson 4
6.RP.2	Understand the concept of a unit rate a/b associated with a ratio $a:b$ with $b \neq 0$ (b not equal to zero), and use rate language in the context of a ratio relationship.	Mission 2, Topic C, Lesson 10 Mission 3, Topic A, Lesson 1 Mission 3, Topic C, Lesson 5-7 Mission 9, Topic A, Lesson 6
6.RP.3	Use ratio and rate reasoning to solve real-world and mathematical problems. a. Make tables of equivalent ratios relating quantities with whole-number measurements, find missing values in the tables, and plot the pairs of values on the coordinate plane. Use tables to compare ratios. b. Solve unit rate problems. c. Find a percent of a quantity as a rate per 100. Solve problems that involve finding the whole given a part and the percent, and finding a part of a whole given the percent. d. Use ratio reasoning to convert measurement units; manipulate and transform units appropriately when multiplying or dividing quantities.	Mission 2, Topic C, Lesson 6-10 Mission 2, Topic D, Lesson 11-14 Mission 2, Topic E, Lesson 15-16 Mission 2, Topic F, Lesson 17 Mission 3, Topic B, Lesson 3-4 Mission 3, Topic C, Lesson 5-9 Mission 3, Topic D, Lesson 10-15 Mission 6, Topic D, Lesson 16 Mission 6, Topic E, Lesson 17 Mission 9, Topic A, Lesson 4-6
The Number System		
6.NS.1	Interpret and compute quotients of fractions, and solve word problems involving division of fractions by fractions.	Mission 4, Topic A, Lesson 3 Mission 4, Topic B, Lesson 4-9 Mission 4, Topic C, Lesson 10-11 Mission 4, Topic D, Lesson 12-14 Mission 4, Topic E, Lesson 16-17
6.NS.2	Fluently divide multi-digit numbers using a standard algorithm.	Mission 5, Topic D, Lesson 9-11
6.NS.3	Fluently add, subtract, multiply, and divide multi-digit decimals using a standard algorithm for each operation.	Mission 5, Topic B, Lesson 2-4 Mission 5, Topic C, Lesson 7-8 Mission 5, Topic D, Lesson 12-13 Mission 5, Topic E, Lesson 14-15

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
		Mission 6, Topic A, Lesson 4 Mission 8, Topic C, Lesson 12 Mission 9, Topic A, Lesson 6
6.NS.4	Find the greatest common factor of two whole numbers less than or equal to 100. Use the distributive property to express a sum of two whole numbers 1–100 with a common factor as a multiple of a sum of two whole numbers with no common factor other than 1. Find the least common multiple of two whole numbers less than or equal to 12.	Mission 7, Topic D, Lesson 16-18
6.NS.5	Understand that positive and negative numbers are used together to describe quantities having opposite directions or values. Use positive and negative numbers to represent quantities in real-world contexts, explaining the meaning of 0 in each situation.	Mission 7, Topic A, Lesson 1, 5
6.NS.6	Understand a rational number as a point on the number line. Use number lines and coordinate axes to represent points on a number line and in the coordinate plane with negative number coordinates. a. Recognize opposite signs of numbers as indicating locations on opposite sides of 0 on the number line. Recognize that the opposite of the opposite of a number is the number itself, and that 0 is its own opposite. b. Understand signs of numbers in ordered pairs as indicating locations in quadrants of the coordinate plane. Recognize that when two ordered pairs differ only by signs, the locations of the points are related by reflections across one or both axes. c. Find and position integers and other rational numbers on a horizontal or vertical number line. Find and position pairs of integers and other rational numbers on a coordinate plane.	Mission 7, Topic A, Lesson 1-2, 4, 7 Mission 7, Topic C, Lesson 11-15
6.NS.7	Understand ordering and absolute value of rational numbers. a. Interpret statements of inequality as statements about the relative position of two numbers on a number line. b. Write, interpret, and explain statements of order for rational numbers in real-world contexts. c. Understand the absolute value of a rational number as its distance from 0 on the number	Mission 7, Topic A, Lesson 3-4, 6-7 Mission 7, Topic B, Lesson 8-9 Mission 7, Topic C, Lesson 13

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	line. Interpret absolute value as magnitude for a positive or negative quantity in a real-world situation. d. Distinguish comparisons of absolute value from statements about order.	
6.NS.8	Solve real-world and mathematical problems by graphing points on a coordinate plane. Include use of coordinates and absolute value to find distances between points with the same first coordinate or the same second coordinate.	Mission 7, Topic C, Lesson 11, 13-15
Expressions & Equations		
6.EE.1	Write and evaluate numerical expressions involving whole-number exponents.	Mission 1, Topic F, Lesson 17-18 Mission 6, Topic C, Lesson 12-15
6.EE.2	Write, read, and evaluate expressions in which letters stand for numbers. a. Write expressions that record operations with numbers and with letters standing for numbers. b. Identify parts of an expression using mathematical terms (term, coefficient, sum, difference, product, factor, and quotient); view one or more parts of an expression as a single entity. c. Evaluate expressions given specific values for their variables. Include expressions that arise from formulas in real-world problems. Perform arithmetic operations, including those involving whole-number exponents, in the conventional order (Order of Operations).	Mission 6, Topic B, Lesson 6, 10-11 Mission 6, Topic C, Lesson 9, 14-15 Mission 6, Topic F, Lesson 18
6.EE.3	Apply the properties of operations to generate equivalent expressions.	Mission 6, Topic B, Lesson 10-11
6.EE.4	Identify when two expressions are equivalent.	Mission 5, Topic D, Lesson 13 Mission 6, Topic B, Lesson 8, 10-11
6.EE.5	Understand solving an equation or inequality as a process of answering a question: which values from a specified set, if any, make the equation or inequality true? Use substitution to determine whether a given number in a specified set makes an equation or inequality true.	Mission 6, Topic A, Lesson 2-5 Mission 6, Topic B, Lesson 8 Mission 6, Topic C, Lesson 15 Mission 7, Topic B, Lesson 9-10

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
6.EE.6	Use variables to represent numbers and write expressions when solving a real-world or mathematical problem. Understand that a variable can represent an unknown number, or, depending on the purpose at hand, any number in a specified set.	Mission 6, Topic A, Lesson 1, 3-5 Mission 6, Topic B, Lesson 6-7 Mission 7, Topic B, Lesson 8, 10
6.EE.7	Solve real-world and mathematical problems by writing and solving equations of the form $x + p = q$; $x - p = q$; $px = q$; and $x/p = q$ for cases in which p , q , and x are all nonnegative rational numbers.	Mission 6, Topic A, Lesson 3-5 Mission 6, Topic B, Lesson 7
6.EE.8	Write an inequality of the form $x > c$, $x \geq c$, $x \leq c$, or $x < c$ to represent a constraint or condition in a real-world or mathematical problem. Recognize that inequalities of these forms have infinitely many solutions; represent solutions of such inequalities on a number line.	Mission 7, Topic B, Lesson 8-10
6.EE.9	Use variables to represent two quantities in a real-world problem that change in relationship to one another. Given a verbal context and an equation, identify the dependent variable, in terms of the other quantity, thought of as the independent variable. Analyze the relationship between the dependent and independent variables using graphs and tables, and relate these to the equation.	Mission 6, Topic D, Lesson 16-18
Geometry		
6.G.1	Find area of triangles, trapezoids, and other polygons by composing into rectangles or decomposing into triangles and quadrilaterals. Apply these techniques in the context of solving real-world and mathematical problems.	Mission 1, Topic A, Lesson 2-3 Mission 1, Topic B, Lesson 4-6 Mission 1, Topic C, Lesson 7-10 Mission 1, Topic D, Lesson 11 Mission 1, Topic G, Lesson 19 Mission 4, Topic D, Lesson 14
6.G.2	Find volumes of right rectangular prisms with fractional edge lengths in the context of solving real-world and mathematical problems.	Mission 1, Topic E, Lesson 15 Mission 4, Topic D, Lesson 14-15 Mission 4, Topic E, Lesson 17
6.G.3	Draw polygons in the coordinate plane given coordinates for the vertices. Use coordinates to find the length of a side joining points with the same first	Mission 7, Topic C, Lesson 15

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
	coordinate or the same second coordinate. Apply these techniques in the context of solving real-world and mathematical problems.	
6.G.4	Represent three-dimensional figures using nets made up of rectangles and triangles, and use the nets to find the surface area of these figures. Apply these techniques in the context of solving real-world and mathematical problems.	Mission 1, Topic E, Lesson 12-16 Mission 1, Topic F, Lesson 18 Mission 1, Topic G, Lesson 19
6.G.5	Use area and volume models to explain perfect squares and perfect cubes.	
Statistics & Probability		
6.SP.1	a. Recognize that a statistical question is one that anticipates variability in the data related to the question and accounts for it in the answers. b. Understand that statistics can be used to gain information about a population by examining a sample of the population; generalizations about a population from a sample are valid only if the sample is representative of that population. c. Understand that the method and sample size used to collect data for a particular question is intended to reduce the difference between a population and a sample taken from the population so valid inferences can be drawn about the population. Generate multiple samples (or simulated samples) of the same size to recognize the variation in estimates or predictions.	Mission 8, Topic A, Lesson 2 Mission 8, Topic B, Lesson 3, 6-7 Mission 8, Topic D, Lesson 17 Grade 7 Mission 8, Topic C, Lesson 12-14 Mission 8, Topic D, Lesson 15-17 Mission 8, Topic E, Lesson 20
6.SP.2	Understand that a set of quantitative data collected to answer a statistical question has a distribution which can be described by its center, spread, and overall shape.	Mission 8, Topic B, Lesson 4-5, 7-8 Mission 8, Topic C, Lesson 11 Mission 8, Topic E, Lesson 18
6.SP.3	Recognize that a measure of center for a quantitative data set summarizes all of its values with a single number while a measure of variation describes how its values vary with a single number.	Mission 8, Topic B, Lesson 6 Mission 8, Topic C, Lesson 9-11
6.SP.4	Display quantitative data in plots on a number line, including dot plots, and histograms.	Mission 8, Topic B, Lesson 3-8 Mission 8, Topic D, Lesson 16-17

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
6th Grade Standards		Lessons
6.SP.5	Summarize quantitative data sets in relation to their context. a. Report the number of observations. b. Describe the nature of the attribute under investigation, including how it was measured and its units of measurement. c. Calculate range and measures of center, as well as describe any overall pattern and any striking deviations from the overall pattern with reference to the context in which the data were gathered. d. Relate the range and the choice of measures of center to the shape of the data distribution and the context in which the data were gathered.	Mission 8, Topic A, Lesson 2 Mission 8, Topic B, Lesson 3-7 Mission 8, Topic C, Lesson 9-12 Mission 8, Topic D, Lesson 13-16 Mission 8, Topic E, Lesson 18
6.SP.6	Understand that the probability of a chance event is a number between 0 and 1 inclusive, that expresses the likelihood of the event occurring. Larger numbers indicate greater likelihood. A probability near 0 indicates an unlikely event, a probability around $\frac{1}{2}$ indicates an event that is neither unlikely nor likely, and a probability near 1 indicates a likely event.	Grade 7 Mission 8, Topic A, Lesson 2-6
6.SP.7	Approximate the probability of a simple event by collecting data on the chance process that produces it and observing its long-run relative frequency, and predict the approximate relative frequency given the probability.	Grade 7 Mission 8, Topic A, Lesson 1, 3-6
6.SP.8	Develop a probability model and use it to find probabilities of simple events. Compare probabilities from a model to observed frequencies; if the agreement is not good, explain possible sources of the discrepancy. a. Develop a uniform probability model by assigning equal probability to all outcomes, and use the model to determine probabilities of simple events. b. Develop a probability model (which may not be uniform) by observing frequencies in data generated from a chance process.	Grade 7 Mission 8, Topic A, Lesson 3-6 Mission 8, Topic C, Lesson 1-13 Mission 8, Topic E, Lesson 1-19

7th Grade

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
Ratios & Proportional Relationships		
7.RP.1	Compute unit rates associated with ratios of fractions.	Mission 2, Topic C, Lesson 8 Mission 4, Topic A, Lesson 2-3 Mission 9, Topic B, Lesson 5
7.RP.2	Recognize and represent proportional relationships between quantities. a. Decide whether two quantities are in a proportional relationship. b. Identify the constant of proportionality (unit rate) in tables, graphs, equations, diagrams, and verbal descriptions of proportional relationships. c. Represent a proportional relationship using an equation. d. Explain what a point (x, y) on the graph of a proportional relationship means in terms of the situation, with special attention to the points (0, 0) and (1, r) where r is the unit rate.	Mission 2, Topic A, Lesson 2-3 Mission 2, Topic B, Lesson 4-6 Mission 2, Topic C, Lesson 7-9 Mission 2, Topic D, Lesson 10-13 Mission 2, Topic E, Lesson 14-15 Mission 3, Topic A, Lesson 1, 3, 5 Mission 3, Topic B, Lesson 7 Mission 4, Topic A, Lesson 3-5 Mission 5, Topic C, Lesson 9, 12 Mission 5, Topic D, Lesson 14 Mission 9, Topic A, Lesson 3
7.RP.3	Use proportional relationships to solve multistep ratio and percent problems.	Mission 3, Topic A, Lesson 5 Mission 4, Topic B, Lesson 6-9 Mission 4, Topic C, Lesson 10-15 Mission 4, Topic D, Lesson 16 Mission 9, Topic A, Lesson 1-4 Mission 9, Topic B, Lesson 6, 8 Mission 9, Topic C, Lesson 13
The Number System		
7.NS.1	Apply and extend previous understandings of addition and subtraction to add and subtract rational numbers. Represent addition and subtraction on a horizontal or vertical number line. a. Describe situations in which opposite quantities combine to make 0. b. Understand addition of rational numbers; $p + q$ is the number located a distance $ q $ from p, in the positive or negative direction depending on whether q is positive or negative. Show that a number and its opposite have a sum of 0 (are additive inverses). Interpret sums of rational numbers by describing real-world contexts.	Mission 5, Topic A, Lesson 1 Mission 5, Topic B, Lesson 2-7 Mission 6, Topic D, Lesson 18 Mission 7, Topic B, Lesson 6

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
	c. Understand subtraction of rational numbers as adding the additive inverse, $p - q = p + (-q)$. Show that the distance between two rational numbers on the number line is the absolute value of their difference, and apply this principle in real-world contexts. d. Apply properties of operations as strategies to add and subtract rational numbers.	
7.NS.2	Apply and extend previous understandings of multiplication and division and of fractions to multiply and divide rational numbers. a. Understand that multiplication is extended from fractions to rational numbers by requiring that operations continue to satisfy the properties of operations, particularly the distributive property, leading to products such as $(-1)(-1) = 1$ and the rules for multiplying signed numbers. Interpret products of rational numbers by describing real-world contexts. b. Understand that integers can be divided, provided that the divisor is not zero, and every quotient of integers (with non-zero divisor) is a rational number. If p and q are integers, then $-(p/q) = (-p)/q = p/(-q)$. Interpret quotients of rational numbers by describing real-world contexts. c. Apply properties of operations as strategies to multiply and divide rational numbers. d. Convert a fraction to a decimal using long division; know that the decimal form of a rational number terminates in 0s or eventually repeats.	Mission 4, Topic A, Lesson 5 Mission 5, Topic A, Lesson 1 Mission 5, Topic C, Lesson 8-11 Mission 8, Topic D, Lesson 16 Mission 9, Topic A, Lesson 4
7.NS.3	Solve real-world and mathematical problems involving the four operations with rational numbers.	Mission 5, Topic B, Lesson 7 Mission 5, Topic C, Lesson 12 Mission 5, Topic D, Lesson 13-14 Mission 5, Topic E, Lesson 15-16 Mission 5, Topic F, Lesson 17 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 6
Expression & Equations		
7.EE.1	Add, subtract, factor, and expand linear expressions with rational coefficients by applying the properties of operations.	Mission 6, Topic D, Lesson 18-22 Mission 9, Topic B, Lesson 7

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.EE.2	Understand that rewriting an expression in different forms in real-world and mathematical problems can reveal and explain how the quantities are related.	Mission 6, Topic B, Lesson 12
7.EE.3	Solve multi-step real-world and mathematical problems posed with positive and negative rational numbers in any form (whole numbers, fractions, and decimals), using tools strategically. Apply properties of operations to calculate with numbers in any form; convert between forms as appropriate. Assess the reasonableness of answers using mental computation and estimation strategies.	Mission 3, Topic C, Lesson 11 Mission 5, Topic C, Lesson 12 Mission 5, Topic F, Lesson 17 Mission 6, Topic A, Lesson 2-6 Mission 6, Topic B, Lesson 11-12
7.EE.4	Use variables to represent quantities in a real-world or mathematical problem, and construct simple equations and inequalities to solve problems by reasoning about the quantities. a. Solve word problems leading to equations of the form $px + q = r$ and $p(x + q) = r$, where p , q , and r are rational numbers. Solve equations of these forms fluently. Compare an algebraic solution to an arithmetic solution, identifying the sequence of the operations used in each approach. b. Solve word problems leading to inequalities of the form $px + q > r$, $px + q \geq r$, $px + q \leq r$, or $px + q < r$, where p , q , and r are rational numbers. Graph the solution set of the inequality on the number line and interpret it in the context of the problem.	Mission 5, Topic E, Lesson 15-16 Mission 6, Topic A, Lesson 4-5 Mission 6, Topic B, Lesson 7-12 Mission 6, Topic C, Lesson 13-17 Mission 7, Topic A, Lesson 7 Mission 9, Topic A, Lesson 3 Mission 9, Topic B, Lesson 7
Geometry		
7.G.1	Solve problems involving scale drawings of geometric figures, including computing actual lengths and areas from a scale drawing and reproducing a scale drawing at a different scale.	Mission 1, Topic A, Lesson 1-6 Mission 1, Topic B, Lesson 7-12 Mission 1, Topic C, Lesson 13 Mission 2, Topic A, Lesson 1 Mission 3, Topic B, Lesson 6 Mission 3, Topic C, Lesson 11 Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lesson 13
7.G.2	Draw triangles when given measures of angles and/or sides, noticing when the conditions determine a unique triangle, more than one triangle, or no triangle.	Mission 3, Topic A, Lesson 2 Mission 7, Topic B, Lesson 6-10 Mission 7, Topic D, Lesson 17

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
7th Grade Standards		Lessons
7.G.3	Describe the two-dimensional shapes that result from slicing three-dimensional solids parallel or perpendicular to the base.	Mission 7, Topic C, Lesson 11, 13
7.G.4	Apply the formulas for the area and circumference of a circle to solve problems.	Mission 3, Topic A, Lesson 3-5 Mission 3, Topic B, Lesson 7-9 Mission 3, Topic C, Lesson 10-11 Mission 9, Topic A, Lesson 4 Mission 9, Topic C, Lesson 11-12
7.G.5	Use facts about supplementary, complementary, vertical, and adjacent angles in a multi-step problem to write and solve simple equations for an unknown angle in a figure.	Mission 7, Topic A, Lesson 2-5
7.G.6	Solve real-world and mathematical problems involving area of two-dimensional objects composed of triangles and trapezoids. Solve surface area problems involving right prisms and right pyramids composed of triangles and trapezoids. Find the volume of right triangular prisms, and solve volume problems involving three-dimensional objects composed of right rectangular prisms.	Mission 1, Topic A, Lesson 6 Mission 2, Topic C, Lesson 8 Mission 3, Topic B, Lesson 6 Mission 7, Topic C, Lesson 12-16 Mission 9, Topic A, Lesson 4 Mission 9, Topic B, Lesson 5, 9
Statistics & Probability		
7.SP.1	Construct and interpret box-plots, find the interquartile range, and determine if a data point is an outlier.	Grade 6 Mission 8, Topic D, Lesson 16-17
7.SP.3	Informally assess the degree of visual overlap of two quantitative data distributions.	Mission 8, Topic C, Lesson 11 Mission 8, Topic D, Lesson 18
7.SP.5	Use measures of center and measures of variability for quantitative data from random samples or populations to draw informal comparative inferences about the populations.	Mission 8, Topic D, Lesson 15-19 Mission 8, Topic E, Lesson 20 Mission 9, Topic A, Lesson 3
7.SP.6	Find probabilities of compound events using organized lists, sample space tables, tree diagrams, and simulation. a. Understand that, just as with simple events, the probability of a compound event is the fraction of outcomes in the sample space for which	Mission 8, Topic A, Lesson 6 Mission 8, Topic B, Lesson 7-10

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS	ZEARN MATH
7th Grade Standards	Lessons
the compound event occurs. b. Represent sample spaces for compound events using methods such as organized lists, sample space tables, and tree diagrams. For an event described in everyday language, identify the outcomes in the sample space which compose the event. c. Design and use a simulation to generate frequencies for compound events.	

8th Grade

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
The Number System		
8.NS.1	Understand informally that every number has a decimal expansion; for rational numbers show that the decimal expansion eventually repeats. Know that other numbers that are not rational are called irrational.	Mission 8, Topic E, Lesson 14-15
8.NS.2	Use rational approximations of irrational numbers to compare the size of irrational numbers, locate them approximately on a number line, and estimate the value of expressions.	Mission 8, Topic A, Lesson 1 Mission 8, Topic B, Lesson 4-5 Mission 8, Topic D, Lesson 12-13
Expressions & Equations		
8.EE.1	Know and apply the properties of integer exponents to generate equivalent numerical expressions.	Mission 7, Topic B, Lesson 2-8 Mission 7, Topic C, Lesson 11, 14
8.EE.2	Use square root and cube root symbols to represent solutions to equations of the form $x^2 = p$ and $x^3 = p$, where p is a positive rational number. Know square roots of perfect squares up to 225 and cube roots of perfect cubes up to 125. Know that the square root of a non-perfect square is irrational.	Mission 8, Topic B, Lesson 2-5 Mission 8, Topic C, Lesson 10 Mission 8, Topic D, Lesson 12-13
8.EE.3	Use numbers expressed in the form of a single digit times an integer power of 10 to estimate very large or very small quantities, and to express how many times as much one is than the other.	Mission 7, Topic C, Lesson 9-12, 14 Mission 7, Topic D, Lesson 16
8.EE.4	Perform multiplication and division with numbers expressed in scientific notation, including problems where both standard decimal form and scientific notation are used. Use scientific notation and choose units of appropriate size for measurements of very large or very small quantities. Interpret scientific notation that has been generated by technology.	Mission 7, Topic C, Lesson 10-15 Mission 7, Topic D, Lesson 16

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
8.EE.5	Graph proportional relationships, interpreting the unit rate as the slope of the graph. Compare two different proportional relationships represented in different ways.	Mission 3, Topic A, Lesson 2-4 Mission 3, Topic B, Lesson 6
8.EE.6	Use similar triangles to explain why the slope m is the same between any two distinct points on a non-vertical line in the coordinate plane; derive the equation $y = mx$ for a line through the origin and the equation $y = mx + b$ for a line intercepting the vertical axis at b .	Mission 2, Topic C, Lesson 10-12 Mission 3, Topic B, Lesson 7 Mission 3, Topic C, Lesson 10-11 Mission 3, Topic E, Lesson 14
8.EE.7	Solve linear equations in one variable. a. Recognize when linear equations in one variable have one solution, infinitely many solutions, or no solutions. Give examples and show which of these possibilities is the case by successively transforming the given equation into simpler forms. b. Solve linear equations with rational number coefficients, including equations whose solutions require expanding expressions using the distributive property and combining like terms.	Mission 4, Topic B, Lesson 3-9
8.EE.8	Analyze and solve pairs of simultaneous linear equations. a. Understand that solutions to a system of two linear equations in two variables correspond to points of intersection of their graphs, because points of intersection satisfy both equations simultaneously. Recognize when the system has one solution, no solution, or infinitely many solutions. b. Solve systems of two linear equations in two variables with integer coefficients: graphically, numerically using a table, and algebraically. Solve simple cases by inspection. c. Solve real-world and mathematical problems involving systems of two linear equations in two variables with integer coefficients.	Mission 3, Topic C, Lesson 13-14 Mission 4, Topic B, Lesson 9 Mission 4, Topic C, Lesson 10-15 Mission 4, Topic D, Lesson 16
Functions		
8.F.1	Understand that a function is a rule that assigns to each input exactly one output. The graph of a function is the set of ordered pairs consisting of an input and the corresponding output.	Mission 5, Topic A, Lesson 1-2 Mission 5, Topic B, Lesson 3-5 Mission 5, Topic E, Lesson 17 Mission 9, Topic A, Lesson 1

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
8.F.2	Compare properties of two functions each represented in a different way (algebraically, graphically, numerically in tables, or by verbal descriptions).	Mission 5, Topic B, Lesson 7 Mission 5, Topic C, Lesson 8
8.F.3	Interpret the equation $y = mx + b$ as defining a linear function, whose graph is a straight line. Recognize examples of functions that are linear and non-linear.	Mission 5, Topic B, Lesson 4, 7 Mission 5, Topic C, Lesson 8 Mission 5, Topic E, Lesson 18
8.F.4	Construct a function to model a linear relationship between two quantities. Determine the rate of change and initial value of the function from a description of a relationship or from two (x, y) values, including reading these from a table or from a graph. Interpret the rate of change and initial value of a linear function in terms of the situation it models, and in terms of its graph or a table of values.	Mission 5, Topic C, Lesson 8-10 Mission 5, Topic D, Lesson 11
8.F.5	Describe qualitatively the functional relationship between two quantities by analyzing a graph. Sketch a graph that exhibits the qualitative features of a function that has been described in a real-world context.	Mission 5, Topic B, Lesson 5-6 Mission 5, Topic C, Lesson 10
Geometry		
8.G.1	Verify experimentally the properties of rotations, reflections, and translations. a. Verify experimentally lines are mapped to lines, and line segments to line segments of the same length. b. Verify experimentally angles are mapped to angles of the same measure. c. Verify experimentally parallel lines are mapped to parallel lines.	Mission 1, Topic A, Lesson 2-4, 6 Mission 1, Topic B, Lesson 7-10 Mission 1, Topic C, Lesson 11, 13 Mission 1, Topic D, Lesson 14 Mission 3, Topic B, Lesson 8
8.G.2	Know that a two-dimensional figure is congruent to another if the corresponding angles are congruent and the corresponding sides are congruent. Equivalently, two two-dimensional figures are congruent if one is the image of the other after a sequence of rotations, reflections, and translations. Given two congruent figures, describe a sequence that maps the congruence between them on the coordinate plane.	Mission 1, Topic C, Lesson 11-13 Mission 1, Topic D, Lesson 15 Mission 2, Topic B, Lesson 6-7

NEW YORK'S LEARNING STANDARDS FOR MATHEMATICS		ZEARN MATH
8th Grade Standards		Lessons
8.G.3	Describe the effect of dilations, translations, rotations, and reflections on two-dimensional figures using coordinates.	Mission 1, Topic A, Lesson 5-6 Mission 2, Topic A, Lesson 4-5 Mission 2, Topic C, Lesson 12
8.G.4	Know that a two-dimensional figure is similar to another if the corresponding angles are congruent and the corresponding sides are in proportion. Equivalently, two two-dimensional figures are similar if one is the image of the other after a sequence of rotations, reflections, translations, and dilations. Given two similar two-dimensional figures, describe a sequence that maps the similarity between them on the coordinate plane.	Mission 2, Topic B, Lesson 6-7, 9
8.G.5	Use informal arguments to establish facts about the angle sum and exterior angle of triangles, about the angles created when parallel lines are cut by a transversal, and the angle-angle criterion for similarity of triangles.	Mission 1, Topic D, Lesson 14-16 Mission 2, Topic B, Lesson 8 Mission 2, Topic D, Lesson 13 Mission 9, Topic A, Lesson 2
8.G.6	Understand a proof of the Pythagorean Theorem and its converse.	Mission 8, Topic C, Lesson 7, 9
8.G.7	Apply the Pythagorean Theorem to determine unknown side lengths in right triangles in real-world and mathematical problems in two and three dimensions.	Mission 8, Topic C, Lesson 6-8, 10
8.G.8	Apply the Pythagorean Theorem to find the distance between two points in a coordinate system.	Mission 8, Topic C, Lesson 11
8.G.9	Given the formulas for the volume of cones, cylinders, and spheres, solve mathematical and real-world problems.	Mission 5, Topic D, Lesson 13-16 Mission 5, Topic E, Lesson 17-21 Mission 5, Topic F, Lesson 22
Statistics & Probability		
8.SP.1	Construct and interpret scatter plots for bivariate measurement data to investigate patterns of association between two quantities. Describe patterns such as clustering, outliers, positive or negative association, linear association, and nonlinear association.	Mission 6, Topic A, Lesson 1-2 Mission 6, Topic B, Lesson 3-8

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8th Grade Standards		Lessons
8.SP.2	Understand that straight lines are widely used to model relationships between two quantitative variables. For scatter plots that suggest a linear association, informally fit a straight line, and informally assess the model fit by judging the closeness of the data points to the line.	Mission 6, Topic B, Lesson 4-6, 8
8.SP.3	Use the equation of a linear model to solve problems in the context of bivariate measurement data, interpreting the slope and intercept.	Mission 6, Topic B, Lesson 6, 8